Immunological Zero Trust Architecture

Using Somatic Hypermutation to Model Zero Trust Architecture

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Abstract

With the constant evolution of cybersecurity threats, many security options such as certain versions of Zero Trust Architecture (ZTA) are being outpaced. Although requests are continuously verified in ZTA, verification policies themselves typically remain static, limiting their ability to adapt to rapidly evolving threats. While adaptive cybersecurity has been explored broadly, work regarding affinity maturation through somatic hypermutation (SHM) for adaptive verification has been comparatively lacking. *Immunological Zero Trust Architecture* (IZTA) is an immune-system-inspired proof-of-concept simulation written in Java that investigates the feasibility of abstracting affinity maturation into an adaptive verification system. IZTA models antigens and B-cell receptors as normalized feature vectors and measures recognition through cosine similarity. SHM is abstracted as iterative stochastic Gaussian mutation, clonal expansion, and affinity-based selection. Each stochastic process is seeded to ensure deterministic and reproducible experiments. System behavior was analyzed through the logging of metrics including affinity, mutation, differentiation, and population diversity. These metrics were used to characterize baseline system behavior and provide a reference for future parameter-sweep experiments. The simulation converged toward higher-affinity solutions while also exhibiting the diminishing returns characteristic of optimization processes. Analysis revealed emergent behaviors, including phases such as exploration, breakthrough, exploitation, and convergence, as well as a tradeoff between population diversity and optimization. These results suggest that complex optimization behavior can emerge from relatively simple local rules governing mutation and selection. Although IZTA is an abstract simulation rather than a deployable cybersecurity system, it demonstrates the computational feasibility of affinity maturation as an adaptive verification strategy and suggests that immunological principles may provide valuable design inspiration for future cybersecurity systems.

Keywords: Cybersecurity, Zero Trust Architecture, immunological, artificial immune system, adaptive policy verification, somatic hypermutation, affinity maturation, clonal expansion, optimization, emergence

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1. Introduction

1.1 Opening Personal Note

My passion for the immune system began after I got a book about it for Christmas in 7th grade. This interest quickly converged with my hobby of computer science. Every year I would take general computer science classes, but in my 10th and 11th grade I took College Board's pilot cybersecurity courses. Additionally, I participated in my high school's cybersecurity club, and joined multiple cybersecurity competitions. Through these avenues I learned about many cybersecurity concepts ranging from networking to red teaming/blue teaming to Linux, but the concept that stuck out to me the most was Zero Trust Architecture.

ZTA is the concept of "Never trust, always verify" (1, 2, 3). The system never implicitly trusts a user just because they passed a previous verification point; at every step the user is verified. I instantly noticed parallels with the immune system. This observation led to a broader question: could the adaptive strategies used by the immune system inspire a model for ZTA? To hopefully answer this while combining my interests I started working on IZTA: *Immunological Zero Trust Architecture*.

1.2 Problem Statement

As cybersecurity threats become increasingly dynamic, fueled in part by advances in technology such as AI, traditional static defences become less effective against rapidly evolving threats.

A promising way to combat dynamic adversaries is with defensive solutions that are equally adaptive. This is the greatest strength of IZTA, in regular ZTA the verifier stays static, while in IZTA the verifier itself evolves to effectively counter the threat.

The answer to making the defences adaptive, perhaps surprisingly, may lie in our own bodies.

The B cells of our immune system employ an ingenious optimization tactic known as "somatic hypermutation." This is essentially making a group of clones from a parent, forcefully mutating them, then checking to see if the clones have become better or worse at detecting the adversary. The clones that perform worse are destroyed in programmed cell-death. Conversely, the clones that get better become the new parent for the next round of SHM (4, 5, 6).

The mutability of B cells allows the immune system to create a wide repertoire of clones, select the best clone (similar to natural selection), and work off that. Eventually the B-cell clones get good enough at recognizing the adversary that the immune system can (hopefully) protect the body (4, 5, 6).

The core simulation of IZTA is inspired by the process of somatic hypermutation. However, it is important to note that while the simulation attempts to model the core principles of SHM, clonal expansion, and affinity maturation (4, 5, 7); the biology is simplified to make this possible in the first place.

2. Core Model

How the model and its constituent parts actually work.

2.1 Biological Inspiration

Antigen (4, 5): The signature of the pathogen (adversary) that is used to evaluate how well a B cell recognizes it.

B cell (4, 5): A cell that has an initial receptor represented as an n-dimensional vector. How well the receptor matches the antigen determines its affinity, measured on a scale from -1.0 to 1.0, where values closer to 1.0 indicate better recognition. During SHM, B cells are cloned and mutated.

Somatic hypermutation (SHM) (4, 5, 7): The overall process consisting of clonal expansion and affinity maturation. During SHM, clones are produced from a parent B cell, stochastically mutated, and repeatedly selected over multiple iterations to improve affinity.

Clonal expansion (4, 5, 7): The process of producing multiple mutated copies of a parent B cell, creating a diverse population of candidate receptors (and consequently, affinity values), for use in SHM.

Affinity maturation (4, 5, 7): The iterative process of improving antigen recognition by repeatedly selecting the highest-affinity clone to become the parent for the next round of SHM.

Plasma cells (4, 5): Differentiated B cells responsible for actively responding to the current adversary. In IZTA, plasma differentiation occurs during the earlier stages of the simulation when an affinity threshold is met. (Simplified from biology.)

Memory cells (4, 5): Differentiated B cells that preserve successful receptors for future encounters with the same adversary. In IZTA, memory differentiation occurs during the later stages of the simulation once an affinity threshold is met. (Simplified from biology.)

T cell (4, 5): Represents the decision layer of the model. It determines whether the current parent or the best clone continues to the next iteration, and whether selected B cells differentiate into plasma cells, memory cells, or remain undifferentiated.

These specific mechanisms are modelled because they directly contribute to adaptive detection and iterative verification, making them the most relevant components for modeling the relationship between immunology and Zero Trust Architecture.

2.2 Mathematical Representation

The principal mathematical objects used in IZTA are vectors. The initial antigen and all B-cell receptors are represented as n-dimensional normalized feature vectors (where n is mutable for experimentation). Representing the antigen and receptors as vectors allows their similarity to be measured using cosine similarity (8).

This was chosen to evaluate the similarity because it measures directional similarity of the vectors independently of vector magnitude. This property makes it well suited for representing receptor-antigen matching as relative feature alignment is more important than absolute values. The resulting value is the affinity score of the receptor to the antigen. Values range from -1.0 to 1.0 where 1.0 represents identical directions and -1.0 represents opposite directions. Vectors are renormalized after each mutation so that affinity depends only on directional similarity rather than changes in vector magnitude. This prevents repeated mutations from artificially inflating vector length over successive iterations.

Mutation is modeled using normally distributed random variables (Gaussian deviates). When mutating a group of clones or creating the initial population of B cell parents, Gaussian deviates are generated and added to the receptor to mimic random mutations where bigger changes are of greater rarity.

Finally, IZTA computes summary statistics, including mean, variance, standard deviation, and iteration-to-iteration change in affinity, to support analysis of convergence and overall simulation behavior.

2.3 Somatic Hypermutation

The essence of the simulation is the SHM algorithm. It begins by creating an initial population of B cells with a variety of different receptors (using Gaussian deviates) which are then scored, using cosine similarity, against the initial antigen. The cell with the highest affinity score is then selected to be the parent and baseline for the next step. From the parent, n clones are created. Each clone has an independent probability for each entry of its receptor vector to mutate.

In the simulation, mutations are modeled by adding a Gaussian deviate (larger mutations are less likely) to the corresponding vector entry. Some mutations are designated as lethal, representing catastrophic receptor changes. These clones receive an affinity score of –1.0 and are excluded from future selection as well as mean-affinity calculations. After the clones are mutated, scores are recomputed and the highest affinity clone becomes the new parent for the next iteration.

2.4 Differentiation

During the middle stage of affinity maturation, randomly selected B-cell clones whose affinity exceeds the plasma threshold may differentiate into plasma cells. During the later stage of the simulation, an analogous process produces memory cells using a separate affinity threshold. These differentiation rules are simplified abstractions intended to approximate the timing of plasma and memory cell formation rather than reproduce the full biological process.

2.5 Reproducibility

The simulation was made deterministic to ensure reproducible results. All instances of random events based on random numbers use a seed to make the random values deterministic. There is a wide variety of configurable parameters to easily reproduce past results while also allowing for vast experimentation and analysis. Because every random process derives from the same seeded pseudorandom number generator, identical parameters always produce identical simulation outputs. Furthermore, all important data is logged and can be exported as a CSV file to make it even more reproducible. The logged data in CSV format lets analysis be more effective and efficient through the ability to use spreadsheets with it.

2.6 SHM Pseudocode

SHM Pipeline:

```
Initialize antigen
Initialize Bcell population
Evaluate affinity

for iteration = 1 to 50
    Make mutated clones from the parent
    Remove dead clones (from lethal mutations)
    Compute and select highest affinity clone
    Differentiate plasma and memory cells
    Log iteration statistics
End
Export CSV
```

2.7 Mutation Code Snippet

Mutation Code:

```
if (rng.nextDouble() < Sim.MUTATION_CHANCE) {
                Mutated = true;
                double mutation = rng.nextGaussian() *
MUTATION_MAGNITUDE;
                receptor[i] += mutation;
            }
if (Mutated && rng.nextDouble() < Sim.LETHAL_MUTATION_CHANCE) {
            clone.setAffinity(-1.0);
```

3. Zero Trust Architecture & Immunology

3.1 ZTA Overview

Zero Trust Architecture (ZTA) is a cybersecurity framework built upon the principle of "Never trust, always verify." Unlike traditional perimeter-based defense, Zero Trust Architecture is applied throughout the system. Zero trust assumes that no user, device, or request is implicitly trusted, regardless if it originated inside or outside the network. Every access request must be continuously evaluated based on available evidence before access is granted (1, 2, 3).

3.2 Immunology mapped to ZTA

Although IZTA is inspired by immunology, its purpose is not to faithfully replicate the immune system, rather, biological components are abstracted into computational analogues. Each biological component either corresponds to a function or adds upon a function commonly found in ZTA.

Table 1

Mapping between immunology, IZTA, and Zero Trust Architecture

Immunologically	**IZTA**	**ZTA**
Antigen	Pathogen signature vector	User, device, or request under evaluation

B-cell receptor	Feature vector	Security policy/Trust evaluator
Affinity	Cosine similarity score	Degree of confidence
Somatic Hypermutation	Stochastic optimization	Adaptive policy refinement [Not common in ZTA]
Clonal expansion	Candidate detectors	[No real analogy]
Affinity maturation	Iterative selection	Improvement of the adaptive policy
Plasma cells	Immediate response	Active enforcement [Not part of ZTA directly]
Memory cells	Long-term storage	Threat intelligence (learned knowledge)
T cell	Decision layer	[No real analogy]

Rather than assigning trust as an immutable property, IZTA repeatedly tests candidate receptors against the current threat signature. Every iteration requires the current parent to compete against newly generated variants, and only the highest-performing candidate continues. This reflects ZTA principle that trust must be reevaluated every time trust is considered. Likewise, plasma and memory differentiation represent two complementary objectives: responding to the current threat while preserving information that may improve responses to future encounters.

3.3 “Never Trust, Always Verify”

The defining principle of ZTA is “Never trust, always verify.” IZTA follows this principle by ensuring that no B cell retains its status solely because it was successful previously. At every iteration the current parent competes with its mutated clones. Every candidate is rescored against the parent, and only the highest-affinity clone survives to become the next parent. Even previously successful cells may be replaced if a better-performing clone emerges. In this way, trust is continuously re-established rather than inherited.

3.4 How does IZTA Improve Cybersecurity?

While most of the biology has ZTA counterparts, some connections are stronger than others (take affinity and clonal expansion for example). And of the biology that maps to ZTA effectively, some of it, namely SHM, is not common in ZTA, if it exists at all.

A key distinction of IZTA is that it continuously improves not only the outcome of verification but also the verification mechanism itself. Traditional Zero Trust systems repeatedly evaluate requests using largely static policies. IZTA instead evolves its detection policy through iterative optimization, allowing the verifier itself to adapt over time.

In most (or all) implementations of ZTA there is no adaptive policy refinement; the policy stays static. Rather than relying on a single fixed policy, IZTA continuously generates nearby policy variants through stochastic mutation, evaluates their performance, and retains only the highest-performing candidate. This creates an optimization loop capable of adapting to changing threats.

This is the main advantage of IZTA, threats that change rapidly can quickly escape detection, but by allowing the policy itself to evolve, dynamic threats can be countered (9).

Threat actors can be equated to pathogens, and cybersecurity can be equated to the immune system. Threat actors, just like pathogens, are constantly changing to evade detection. The way the immune system counteracts this is by also changing, by adapting. If cybersecurity could become more dynamic, threat actors may be detected more effectively, malware can be more effectively detected.

While this project is focusing on how immunology can further ZTA, immunology can also be used for other cybersecurity concepts. Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) may benefit greatly from incorporating adaptive mechanisms inspired by the immune system. Furthermore, IDS and IPS may mirror and benefit from immunology even more than ZTA because the primary purpose of IDS and IPS closely resembles that of the immune system: detecting and responding to threats. Adaptive IDS and IPS systems could potentially detect a broader range of evolving threats by

incorporating immune-inspired mechanisms such as continual refinement and iterative selection.

3.5 Limitations of IZTA

Like any computational model, IZTA contains numerous simplifications and assumptions. Many are intentional design decisions made to balance biological realism with computational tractability.

Simplified Biology: The biology in IZTA is greatly simplified. SHM in reality is a very complex process consisting of multiple interacting components such as Follicular Dendritic Cells (FDC), T cells, B cells, antigens, cytokines, germinal centers, and more.

Lethal Mutations: Take the lethal mutation, in the simulation it "kills" the B cell (setting its affinity to -1.0), but in reality the B cell goes to a FDC and tries to grab the antigen the FDC is holding, but since the receptor encountered a lethal mutation, it is unable to do so. When the B cell tries to present the antigen to the T cell it receives no survival signals, eventually dying of neglect.

Differentiation: The selection of a B cell to become a plasma or memory cell is much more complex than just checking if its affinity is good enough, in reality it depends on a complex network of signaling molecules, transcription factors, and cellular interactions rather than affinity alone.

The Environment: The environment in IZTA only has 1 antigen that is being optimized for, but in the real immune system many antigens are responded to at once. The real immune system also consists of trillions of cells and interacting parts, and due to the fact that the body is extremely large while cells are extremely small, everything occurs through random (brownian) motion. Since molecular interactions rely on random diffusion, productive encounters between specific immune cells are rare (this is the reason why it takes ~1 week for the adaptive immune system to become activated fully).

Static Threat Representation: Antigens are part of larger pathogens that have extremely fast reproductive cycles and mutate constantly. Due to natural selection, these pathogens

evolve quickly; consequently, their corresponding antigens change rather than remaining static as they do in IZTA.

Vector Representation: While vectors allow for flexibility and access to cosine similarity, it is nowhere near as complex as molecular bindings.

This is just a Simulation: IZTA hasn't been integrated into real-world cybersecurity systems. It hasn't been tested against anything to see if it would actually work in cybersecurity contexts. Currently IZTA is best for analysis of how changing certain parameters affect the larger picture of affinity maturation.

IZTA is not 1-1 with actual cybersecurity, improvements in the simulation may not correspond with expected improvements if implemented in the real world. For example, higher affinity does not imply better malware detection and prevention. While ideally it would, since IZTA is solely a simulation, it is unable to be tested.

Optimization versus Guarantees: IZTA is fundamentally an optimization model. While it attempts to improve detection through iterative refinement, increasing affinity within the simulation does not necessarily translate to improved detection accuracy in real-world cybersecurity systems. Additional validation would be required before such conclusions could be drawn.

That being said, I plan on addressing many of these limitations, such as static threat representation, the environment, and it being just a simulation. Implementing antigen mutation shouldn't be too difficult as I already have implemented receptor mutation. To make the environment more accurate I can increase the scale of everything while decreasing probabilities, and add more antigens. To bring IZTA from being a simulation to actually testing it in the real world, future work could include collaboration with researchers or cybersecurity practitioners to evaluate IZTA on realistic security datasets.

4. Data Analysis

This section examines the behavior of IZTA under baseline conditions and through parameter variation. The baseline configuration was designed to approximate the intended

biological behavior while maintaining computational efficiency. Simulation parameters are recorded once at the beginning of each run, while iteration-specific metrics are logged after every iteration. Refer to the appendices for details.

System behavior and sensitivity is the main thing being analyzed. I looked for patterns across multiple metric families and their graphs to get a better understanding of how the immune system works.

The goal of this data analysis was to understand the behavior of the system: convergence, parameter sensitivity, and emergent behaviors, while looking for any ZTA implications.

4.1 Baseline System Analysis

4.1.1 Baseline Parameters

Refer to table 2 for specific baseline details.

4.1.2 *Baseline Dynamics Graphed*

Basic Metrics

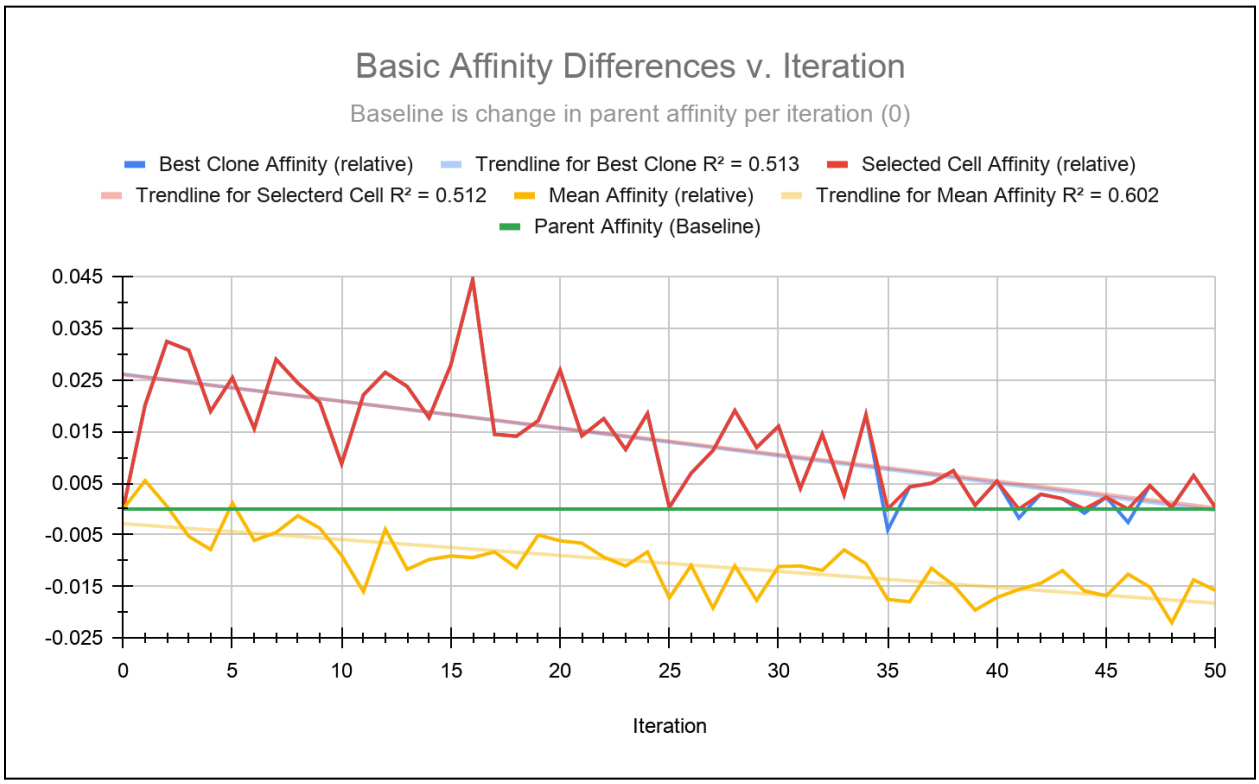


Figure 1. Relative affinity changes of the best clone, the selected cell, and the population mean normalized against the parent affinity baseline over 50 iterations.

This graph compares the affinity of the best clone, the selected cell, and the mean clone affinity relative to the parent cell throughout the simulation. Parent affinity is normalized to zero so that all other values represent changes relative to the current parent rather than absolute affinity.

The selected cell (red) is the cell that advances to the next iteration. Because the parent is retained when every clone is worse than the parent the selected affinity is always greater than or equal to the parent affinity (0). The best clone (blue) represents only the highest-affinity clone produced during clonal expansion. In most iterations the selected cell and best clone are identical; however, when every clone performs worse than the parent, the parent is retained, causing the two curves to diverge. These divergence points correspond to iterations in which no beneficial mutation was produced. Such points

include: iteration 35, -4.06×10^{-3}; Iteration 41, -1.73×10^{-3}; iteration 44, -8.01×10^{-4}; iteration 46, -2.59×10^{-3}.

The mean affinity (yellow) remains consistently below (except for iterations 1, 2, and 5) the parent affinity because most random mutations are either neutral or detrimental to antigen recognition. This reflects a fundamental property of somatic hypermutation: beneficial mutations are rare, and improvement occurs through selection rather than because mutations are inherently advantageous.

At iterations 1, 2, and 5 mean affinity is at 5.50×10^{-3}, 5.51×10^{-4}, and 1.18×10^{-3} respectively. These iterations being the only iterations with a positive mean affinity value may just be due to chance, but further testing is needed to learn the actual cause. After iteration 5 mean affinity continues decreasing to -1.57×10^{-2}, with the lowest at iteration 48 with -2.20×10^{-2} affinity gain. Iteration 5 being positive implies that the entire repertoire of clones were better than the parent. While this trend initially suggests that they are close to each other in affinity value because the affinity range and standard deviation are low at iteration 5, iterations 1 and 2 have a very high affinity range and standard deviation (as seen in the population diversity graphs) despite also being positive.

Consequently, affinity maturation is driven not by the average clone improving, but by the repeated selection of rare high-performing variants. This is further supported by the gradual downward trend in mean affinity throughout the simulation (slope of -3.14×10^{-4}). Although the mean affinity fluctuates from iteration to iteration, often mirroring changes in the best clone and selected cell, it steadily decreases over time as selection repeatedly promotes only the highest-performing clones while the majority of mutations remain neutral or detrimental.

Both best clone and selected cell trend downwards as the iterations progress. This behavior is expected because early in the simulation the parent cell is unoptimized, so random mutations can easily cause a big jump (see iteration 16); however, later in the simulation, the parent cell is already very optimized, so the random mutations run into diminishing returns.

In selected cell, the average gain from iterations 1-16 is 2.77×10^{-3}, while the slope from iterations 16-50 is -1.29×10^{-3}, indicating that the rate of exploration is much higher than the rate of exploitation of the most beneficial mutations. While the net displacement per iteration is 2.77×10^{-3} from iterations 1-16, the actual path distance traveled by the system during this phase is vastly higher at 0.388. From iterations 16-50, net displacement per iteration is -1.29×10^{-3}, but total path distance is 0.325. This contrast implies that the algorithm is computationally more efficient when exploiting a beneficial mutation rather than exploring a search space.

The algorithm, just like the immune system, is converging on the optimal affinity. Note iteration 16, this iteration has the highest spike at 0.044 affinity gain from the parent, signifying it as the iteration with the greatest beneficial mutation. This extremely beneficial mutation also has the added effect of pulling up the mean affinity greatly. One highly successful mutation is all the immune system needs to direct an entire generation to the optimal solution.

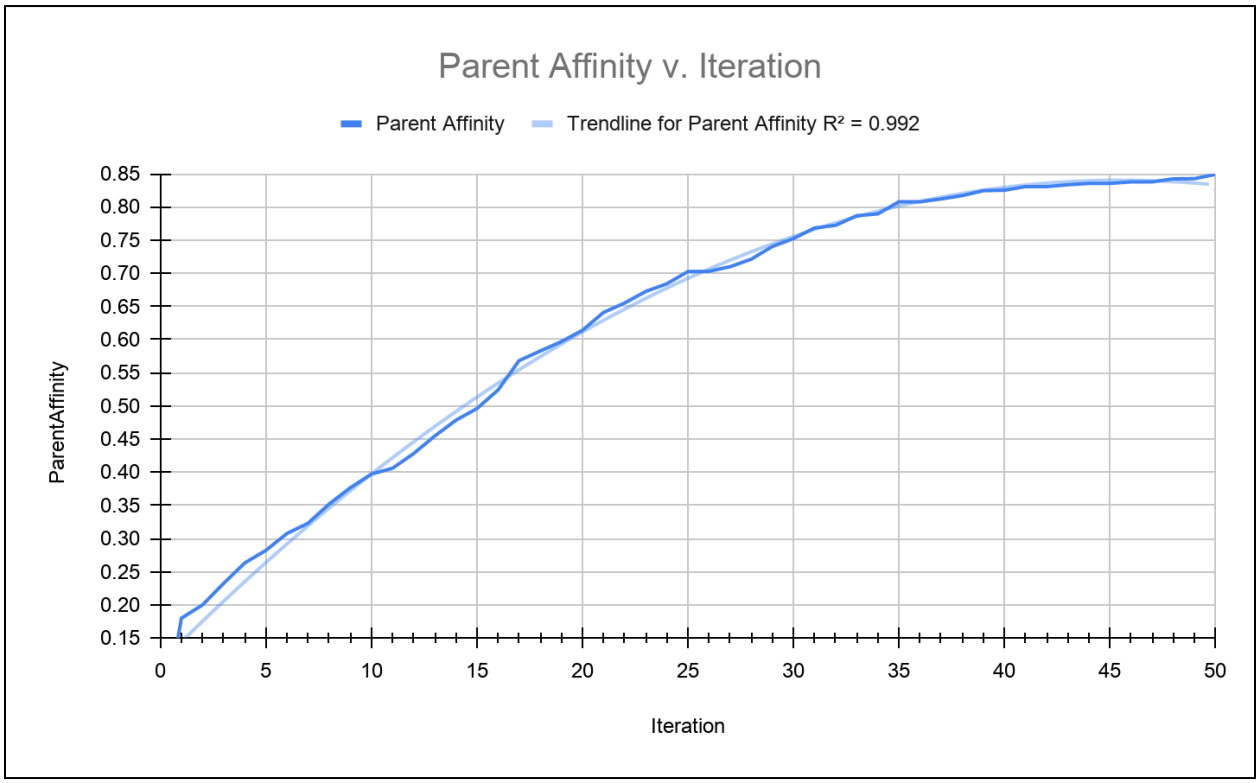


Figure 2. Trajectory of absolute parent affinity growth over 50 iterations fitted with a quadratic polynomial trendline to model sigmoidal optimization deceleration.

This graph is intended to provide the missing information of the first graph, tracking how parent affinity itself changed. Parent affinity increases in a shallow sigmoidal curve (S-curve).

During the first 16 iterations, parent affinity increases at a steady rate (from 0.180 to 0.523), with the biggest jump from iteration 16 to 17 (0.523 to 0.568). This period corresponds to a high exploratory phase of the algorithm, where the parent is still far from the optimal region of the search space and random mutations are more likely to produce substantial gains.

Iteration 17 acts as the transition point for this curve as it is when the rate of increase starts decreasing. This aligns with the concept of diminishing returns: as the parent becomes increasingly optimized, most nearby mutations either keep or slightly worsen performance rather than producing large improvements. This behavior is consistent with the earlier graph, where the rate of affinity gain decreases over time.

In later iterations, the curve begins to plateau, suggesting convergence toward a local optimum. At this stage, the system becomes increasingly sensitive to small variations, and most mutations have minimal impact on affinity. As a result, improvements occur more slowly and primarily through minor micro-adjustments rather than the large, sweeping breakthroughs seen early on.

Parent affinity ranges from 0.18-0.849 which implies an average rate of change of 0.013 but that isn't the complete picture. The slope from iterations 1-16 is 0.022, while the slope from iterations 17-50 is 0.008, aligning with the idea of convergence preceded by exploration rather than consistent increase.

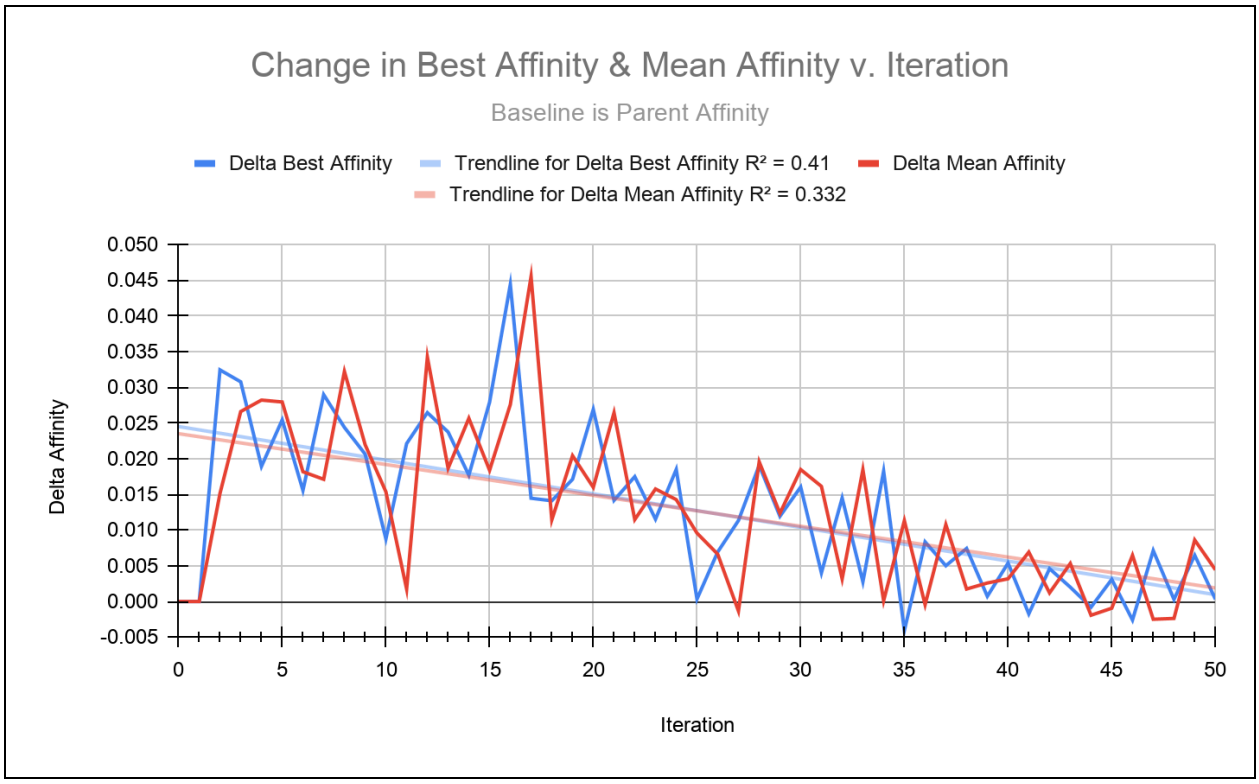


Figure 3. Iteration-to-iteration velocity changes (Delta) in best clone affinity and mean population affinity across the simulation lifecycle.

This graph is meant to show the per-iteration change of the best and average affinity across the simulation lifetime. While much of this data isn't providing any new information, it provides something more important: a new perspective. The perspective this provides is the volatility of the algorithm across iterations.

Over time, both Delta Best Affinity and Delta Mean Affinity decrease greatly. Volatility started high and decreased as iterations passed. The Delta Best Affinity and Delta Mean Affinity are very high earlier in the simulation which is consistent with the iterations before 17 being an exploratory phase. After the peak surge at iterations 16-17 (0.044 best affinity & 0.045 mean affinity), diminishing returns start to take hold as changes from mutations become less likely to greatly increase affinity. At an iteration where all the clones performed worse than the parent (such as iteration 35 at -0.004 best affinity), Delta Best Affinity goes into the negatives, with Delta Mean Affinity following (-4.47×10^{-4} mean affinity).

One interesting pattern that appears is Delta Mean affinity seems to follow Delta Best affinity, only offset by an iteration. This idiosyncratic behavior is caused by how the

algorithm works rather than a biological explanation. When an unusually effective mutation happens, Delta Best spikes instantly due to the mutation making the Best clone much better than the parent. Delta Mean Affinity peaks one iteration later because the breakthrough clone becomes the parent for the following generation. Its descendants therefore begin from a stronger starting point, increasing the average affinity of the next generation rather than the current one.

As both Delta Best Affinity and Delta Mean Affinity start to oscillate around 0, the system appears to approach convergence. Take iterations 40-50, the Delta Affinity is extremely close to 0 and vibrates around 0, signifying that most mutations tend to have a neutral effect, while only a few are beneficial.

Mutation Metrics

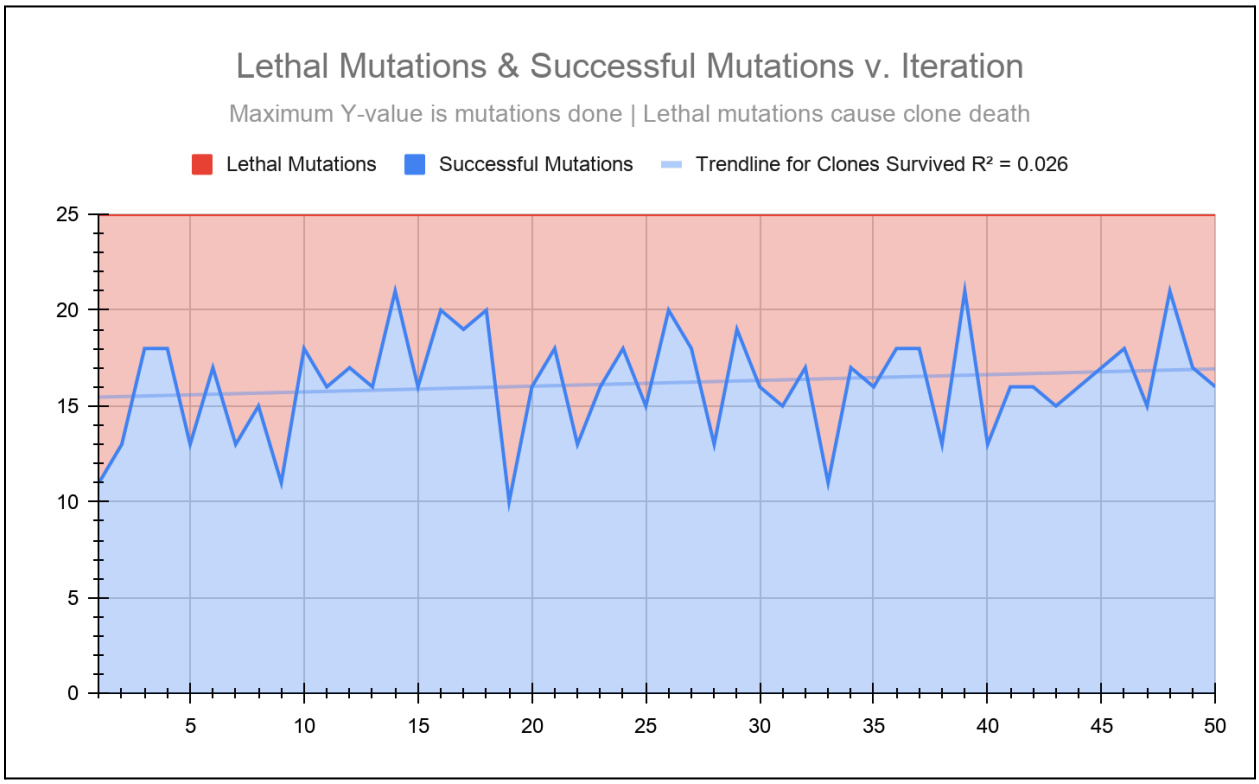


Figure 4. Generational breakdown of the 25-clone population evaluated to track the distribution of successful versus lethal mutations under a static 35% hazard rate.

This graph specifically shows whether a mutation was lethal or not per generation. While the graph does not provide overarching insights, it allows us to see outliers and how it may be reflected in the other graphs.

At every iteration, 25 clones are mutated (as seen in the y-axis), and if a B cell's receptor vector is mutated, then a flat 35% chance for a lethal mutation is applied. On average, 16 cells have no lethal mutations and 9 cells have at least one lethal mutation, which is consistent with the 35% chance for lethal mutations.

While the trendline shows a slightly increasing trend, this is purely due to chance because no aspect of the algorithm affects lethal mutation chance and statistically, the R^2 is only 0.026. And if the P-value is calculated through a t-test then it will become obvious that the trendline being non-zero is due to chance only.

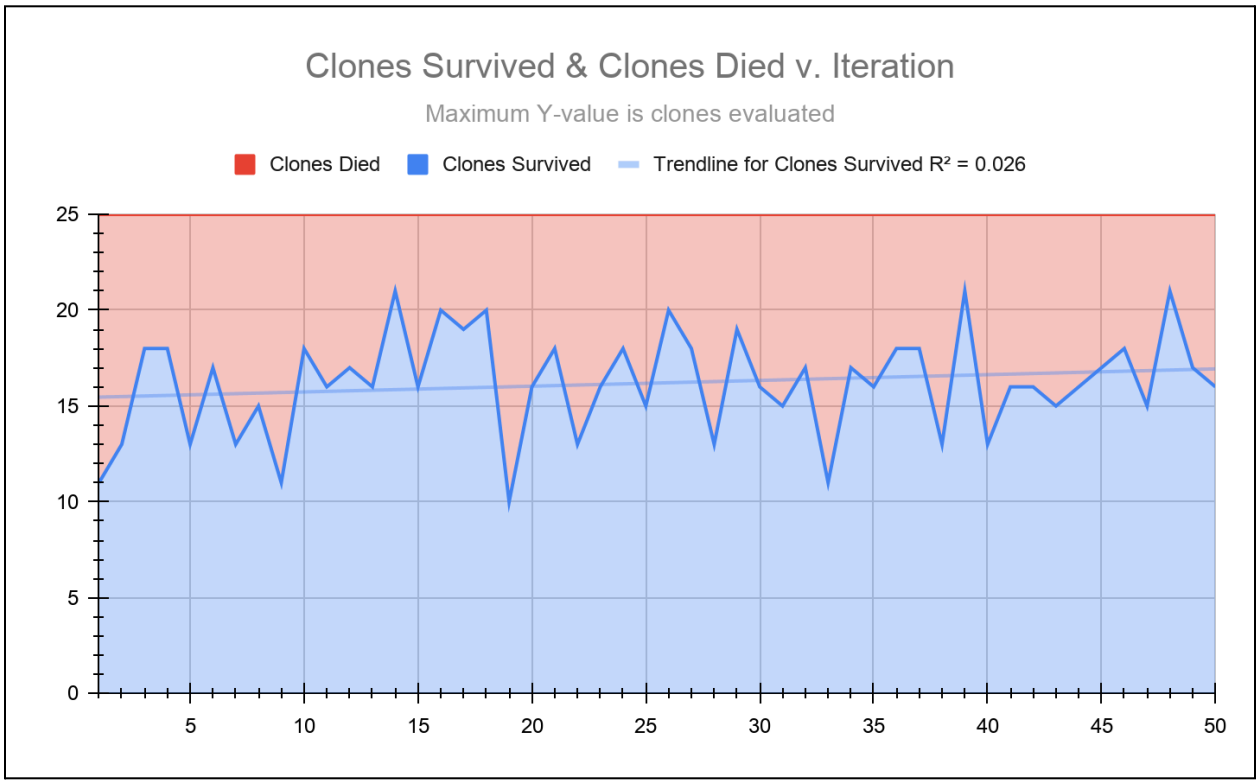


Figure 5. Candidate pool survival dynamics tracking the absolute quantity of surviving versus culled clones available for iterative affinity maturation per generation.

This graph looks at how many clones survived and died out of 25 clones across 50 iterations. This graph may seem similar to the previous one; in fact, they are identical. But this graph makes obvious another use for this data. This graph emphasizes the size of the

surviving candidate pool available for affinity maturation in each generation which may make some trends or outliers easier to characterize.

Due to how the code works, the only death condition for a clone is to encounter a lethal mutation, otherwise the clone lives. Consequently, the number of surviving clones is simply the complement of the number of lethal mutations.

Differentiation Metrics

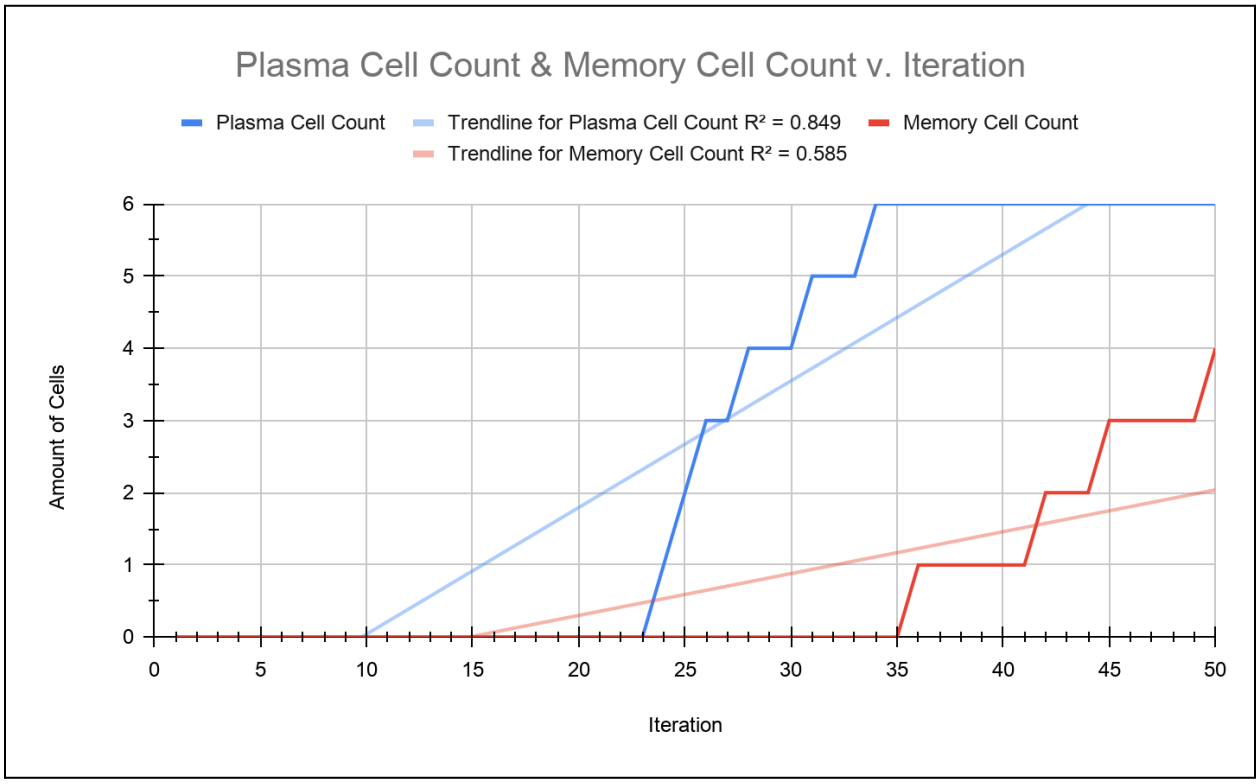


Figure 6. Cellular kinetics modeling the linear accumulation and hard population ceilings of differentiated plasma cells and memory cells over time.

This graph shows the amount of plasma and memory cells across iterations. This graphical representation allows us to visualize the different growth rates and timing of these cells.

Despite the plasma start iteration being at iteration 15, plasma cells are initially produced at iteration 24. This is because for a cell to differentiate into a plasma cell, the cell must also meet the threshold requirement (0.7). Even after iteration 15, affinity hasn't been optimized enough for a cell to be differentiated into a plasma cell. Plasma cell production halts at

iteration 35 because the logic behind differentiation in the simulation specifically has plasma cell production end when memory cells start being made.

The memory start iteration is at iteration 35, and by iteration 36 a memory cell is produced. Although the memory start iteration is later than plasma start iteration, memory threshold is much lower (only 0.5). Since affinity was already optimized above 0.50 by iteration 35, memory cells started being produced quickly after.

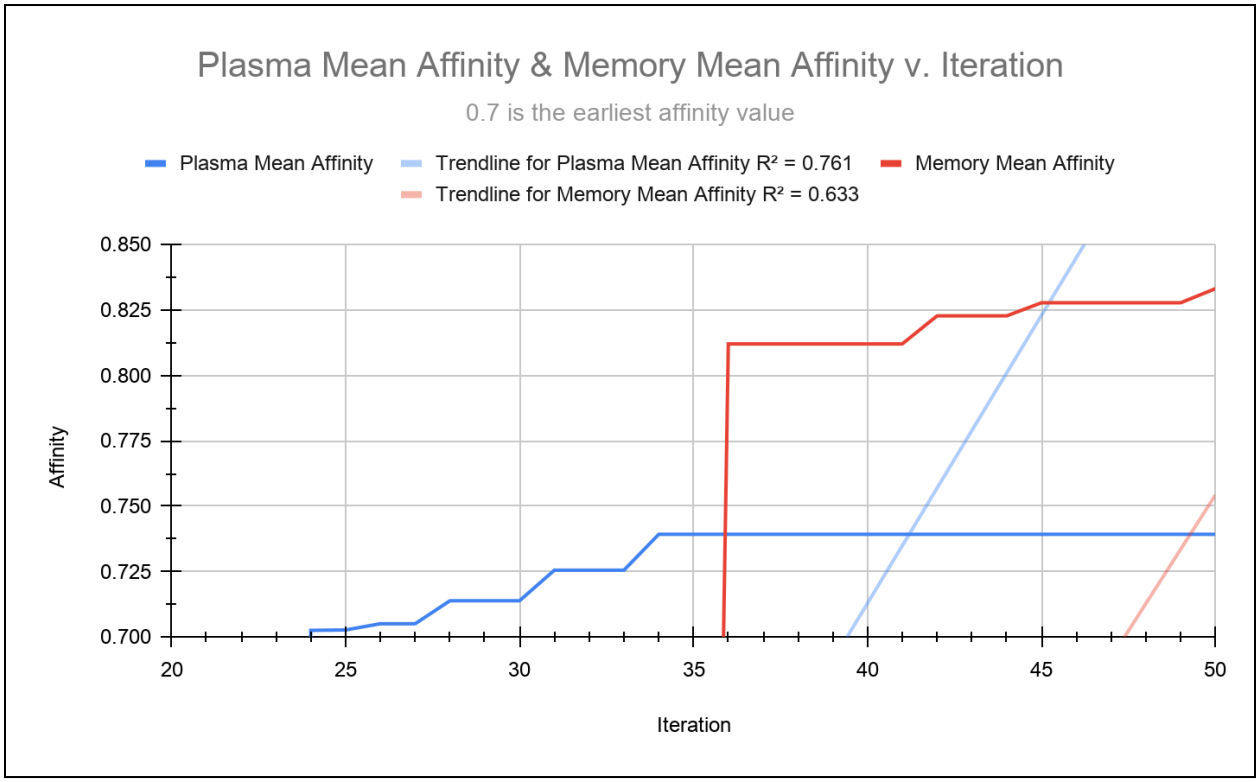


Figure 7. Localized mean affinity trajectory of differentiated plasma and memory cell sub-populations; standard linear trendlines are noted as mathematically unrepresentative of the step-stair data.

This graph is intended to show how mean affinity changes for both plasma and memory cells. In conjunction with the previous graph, this provides a full picture on differentiation: the amount and effect of these cells.

Plasma mean affinity appears at iteration 24, which is consistent with the previous graph showing the first plasma cell originated in iteration 24. Plasma mean affinity growth is very slow, with it starting at 0.702 and ending at 0.739.

Memory mean affinity starts off at a much higher mean affinity because it occurs later in the simulation, after optimization has already partially occurred.

Surprisingly, there is a relatively long period where the memory mean doesn't change: from iteration 36 to iteration 41. From looking at the previous graph, this is because no new memory cells were added, rather than the new ones having the exact affinity as the mean. The reason for this cannot be because the affinity threshold wasn't passed as parent affinity was higher than 0.5 the entire time (**See Figure 2**). Given this reasoning, the most likely explanation is that no clone passed the 50% chance to differentiate in those 5 iterations, despite it being unlikely (3.125% chance of occurring).

Although very simplified, this ordering qualitatively resembles the biological immune response. Plasma cells are generated earlier to respond to the ongoing infection, whereas memory cells appear later, after affinity maturation has produced receptors with higher average affinity.

Population Diversity Metrics

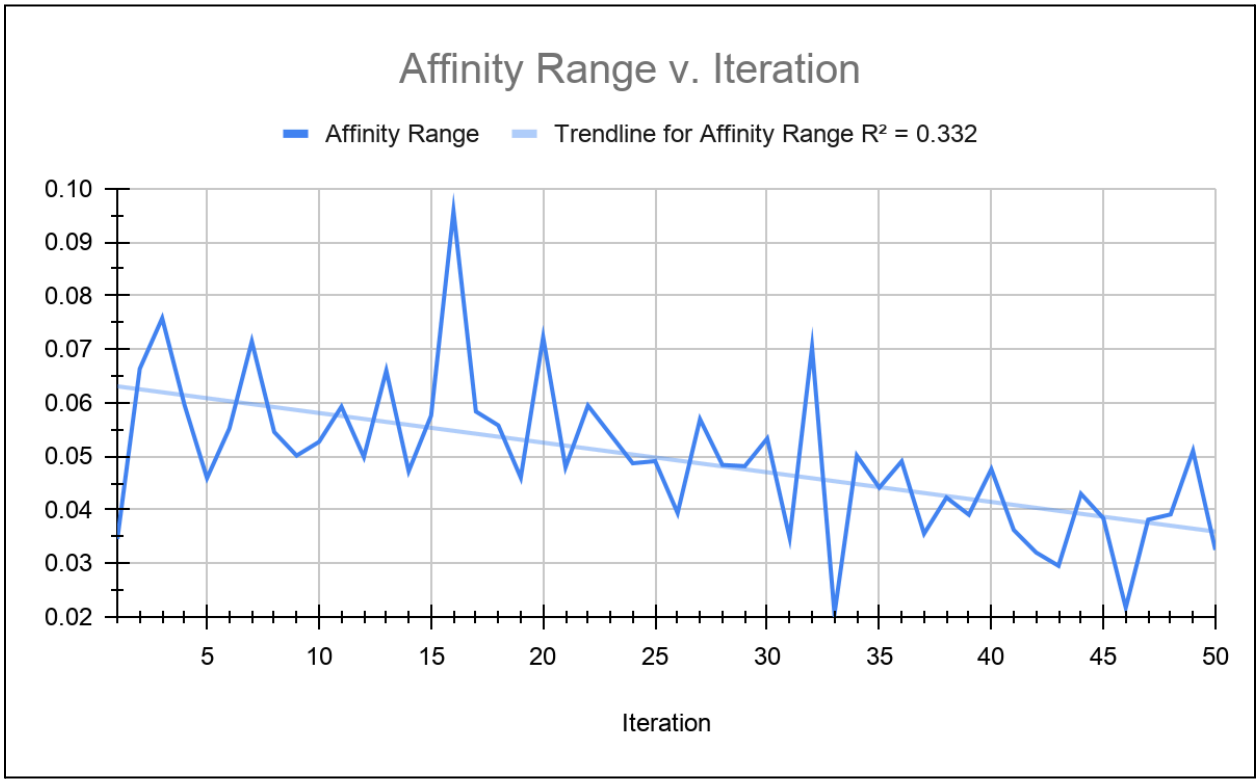


Figure 8. Absolute generational affinity range tracking population affinity range (Max - Min) to isolate lone outlier breakthroughs across the optimization timeline.

This graph is meant to provide insights on how diversity is affected by simulation mechanisms. Additionally, this graph provides further context on certain aspects of other graphs (such as diminishing returns).

Affinity range generally trends downwards across iterations (with the trendline having a slope of -5.53×10^{-4}). This is congruent with the starting iterations being for exploring the search space, the middle iterations for exploiting it, and the last iterations for convergence. At first, the affinity range is high at ~0.07, and actually trends higher until iteration 16 at 0.095. A high-affinity range implies that there is a great difference between the best and worst clones; this happens when affinity is unoptimized. Affinity range lowering indicates that the algorithm is converging on an optimized affinity.

The most noticeable part of this graph is the spike at iteration 16 (0.095). The uptick at iteration 16 is due to a single massively successful mutation causing the best B-cell clone to be much better than the rest of the pack, temporarily increasing the affinity range.

The rise at iteration 32 is not immediately obvious in the other graphs; however, in this graph, iteration 32 sticks out as it causes a great but temporary increase in affinity range. But what is most notable about iteration 32 is iteration 33. Specifically after iteration 32, the biggest dip in affinity range occurs from 0.070 to 0.020 (a difference of 0.049). This implies that iteration 33 had the most similar clones. The spike into a heavy decline seems to be due to pure chance, as both iteration 32 and 33 are outliers, but there may be an unknown underlying reason.

This graph also provides context to the first graph (**See Figure 1**). For the majority of iterations, Best Clone Affinity and Selected Cell Affinity are above 0, while Mean Affinity is below 0. Affinity range is higher when the gap between Best Clone and Selected Cell, and Mean Affinity is greater. At iteration 33, there is a small increase in Mean Affinity while Best Clone and Selected cell both decrease slightly.

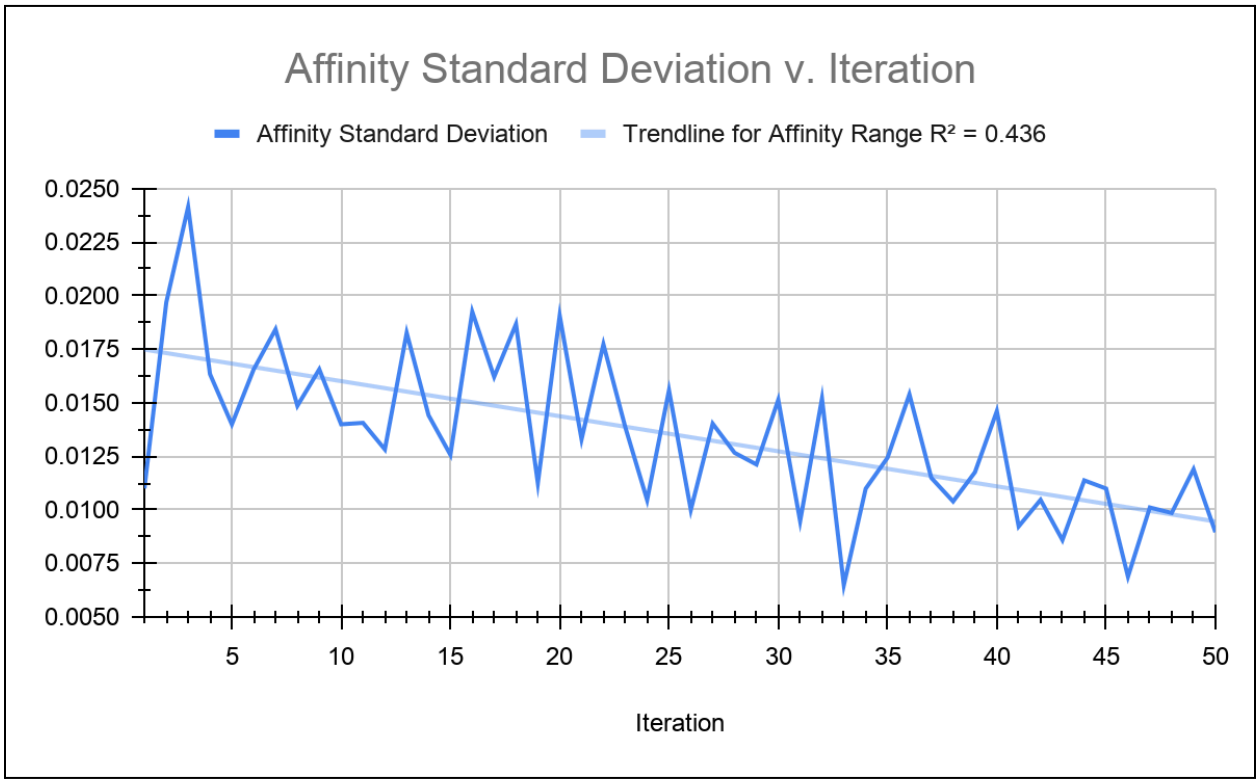


Figure 9. Generational affinity standard deviation measuring overall population dispersal to track system-wide algorithmic convergence.

This graph is meant to complement the previous one by bringing more information to support it. The trend in this graph is very similar to the previous one as Standard Deviation and Range are similar but with key differences.

Affinity standard deviation decreases across iterations, implying the same thing as the previous graph, but with a different perspective. Early in the simulation, variability is high as the affinity standard deviation is highest at iteration 3 (0.024). The B cell and its clones are exploring the search space, and with big jumps from mutations, each clone is highly variable. As the algorithm converges, affinity standard deviation decreases because each clone becomes more like its sister clones as they all approach the local optimal.

Although iteration 16 isn't an outlier in this graph, iteration 33 still is. Iteration 33 is the greatest decrease in standard deviation, which is consistent with Iteration 33 also being the iteration with the lowest affinity range.

One unexpected difference between this graph and the previous one is iterations 10-12. In this graph, iteration 10 and 11 are very similar (0.0139 vs 0.0140), while iteration 12 is a slight decrease from iteration 11. By contrast, in the affinity range graph, 11 is a local maximum between iterations 10 and 12. Seemingly in this graph, the values of iteration 11 and 12 are decreased relative to iteration 10 in the graph of affinity range. This may be because iteration 11's small increase in standard deviation was enough to greatly increase affinity range, and iteration 12's small dip in standard deviation was enough to significantly decrease affinity range.

Another difference is that from iteration 47-48 in this graph, standard deviation decreases, but contrary to intuitive sense, affinity range actually increases. This discrepancy can most likely be attributed to most clones being centered around a single value while the best clone is far ahead of the rest and/or the worst clone is far behind the rest. The overall standard deviation would decrease but affinity range would increase.

4.2 Overall Observations

While I made many graph-specific observations, many patterns I noticed were global and applied to multiple graphs and metrics.

4.2.1 Emergent Phases of Optimization

One of the most interesting types of behavior in any intricate system dictated by multiple local rules is emergent behavior. Complex behavior that emerges through the interaction of relatively simple component behaviors. The emergent behaviors in IZTA were not explicitly coded; rather, they arose naturally from the interaction of simple algorithmic rules.

Exploration (1-15):

This phase occurs in the early stages of the simulation; in the baseline, exploration occurs from iterations 0-15. In exploration, the search space is still novel and mostly unoptimized.

Since starting parent affinity is very low, mathematically, random mutations that are positive can have a greater beneficial effect. Every graph in the basic & population diversity metric families depicts this in one way or another. This is most obvious in **Figure 2**, initially the growth rate is relatively fast, but later on it plateaus.

In **Figure 1** and **Figure 3**, the change in affinity starts off high and decreases due to the mutations being less likely to cause a big change. In both population diversity graphs, diversity is initially high due to both the parent initial variability and clones searching in all directions. As successful variants are selected and propagated, diversity gradually decreases, marking the transition toward exploitation.

Breakthrough (16):

A breakthrough represents a major improvement that shifts the system from broad exploration toward refinement of a successful solution. The exact relationship between breakthrough magnitude and phase transitions requires further testing, but in this baseline run, it's at iteration 16. While the magnitude of a breakthrough may vary greatly, selection still exploits it.

In this run, depicted by multiple graphs (See **Figure 1**, **Figure 2**, **Figure 3, Figure 8**), the breakthrough mutation happened at iteration 16. Iteration 16 had a single, yet massively beneficial mutation that had a host of consequences. It pulled up the mean affinity indirectly, it increased affinity range greatly, and most importantly: it led to its exploitation through selection.

Exploitation (17-35):

This phase occurs in the middle iterations of the simulation and after a breakthrough. The system works to effectively take advantage of the previous breakthrough.

During iterations 17-35, affinity range and affinity standard deviation start decreasing fast. Now, each clone starts nearing the high-affinity region instead of just the best clone of the generation.

At this point, the search space around the current solution is more optimized and now changes are of much lower magnitude, and only work to narrow around the apparent local optimum. Selection becomes more important too as large improvements become rarer. During iterations 17-35, stochastic mutations are still providing benefits but diminishing returns become much more apparent.

Convergence (36-50):

This phase is characterized by the near cessation of any meaningful improvements, which tends to happen later in the simulation. By this point, the search space around the high-affinity region is highly optimized and only miniscule improvements happen. Every graph in the basic & population diversity metric families describes this.

In **Figure 1**, both the best clone and selected cell tend towards 0 relative to the parent. It gets to a point where the selected cell is seemingly oscillating right above the 0 line, while the best clone goes into the negatives a couple times. Additionally, in **Figure 3**, delta also starts to oscillate around 0. In **Figure 2**, parent affinity plateaus in a manner similar to sigmoidal and logarithmic curves.

Since meaningful improvements become increasingly rare, every generation instead becomes more similar because every generation's clones converge on the same local optimum high-affinity state. In both population diversity graphs, affinity range and affinity standard deviation keep decreasing, indicating that each generation is getting more similar.

4.2.2 *Optimization Through Selection*

In general, the majority of mutations tend to be neutral or deleterious. This is clearly seen in **Figure 1**, where mean affinity is consistently lower than that of the parent's. To make use of the beneficial mutations while discarding the other ones, a mechanism is needed to facilitate this.

Selection is exactly that, without selection, random mutations would have an overall negative effect on affinity, but by implementing selective behavior, mutations can be filtered. Without a selective barrier, optimization would not be sustained because deleterious mutations would negate it. By definition, optimization is maximizing an aspect, and the most effective and efficient way is through a simple comparison.

The simulation demonstrates that optimization is driven by the interaction between mutation and selection. Mutation generates diversity and occasionally produces improvements, while selection preserves those improvements and prevents the overall population from drifting toward lower affinity. Without this selective process, affinity maturation would not emerge despite the continual production of new variants.

4.2.3 *Diversity v. Optimization*

Diversity and optimization are both fundamental to the simulation, but their relationship is particularly interesting. Overall, the baseline run suggests a tradeoff between the two. Early in the simulation, diversity is high while optimization is relatively low. As affinity improves over successive generations, diversity gradually decreases as clones become increasingly similar.

This tradeoff creates its own optimization problem: when should the system prioritize diversity, and when should it prioritize optimization? The emergent phases observed in the baseline run provide a possible answer. During exploration, high diversity allows the algorithm to search broadly through the solution space, increasing the likelihood of

discovering highly beneficial mutations. After a breakthrough occurs, selection increasingly favors refinement over exploration, causing diversity to decline as the population converges toward a high-affinity region.

Rather than explicitly programming this balance, it emerged naturally from the interaction between stochastic mutation and selection. The simulation therefore suggests that maintaining high diversity is most valuable early in optimization, whereas reducing diversity becomes advantageous once a solution has been identified.

4.2.4 Summary of Baseline Behavior

Overall, the baseline simulation behaved consistently with how IZTA was designed and coded. Across 50 iterations parent affinity increased steadily while improvements exhibited diminishing returns, indicating convergence towards a high-affinity solution. Although most mutations were neutral or detrimental, the interaction between stochastic mutations and selection enabled optimization.

Several emergent behaviors also surfaced without being explicitly programmed. The simulation naturally exhibits phases corresponding to exploration, transition, exploitation, and convergence, while population diversity decreased as affinity was further optimized. These emergent dynamics closely resemble the qualitative behavior of affinity maturation in the adaptive immune system and provide evidence that relatively simple local rules can produce complex system-wide behavior.

4.3 ZTA Implications

Evolution of Verification Policy: Since traditional ZTA applies a static authenticator (often times a comparison check of log-in credentials), attacks can learn how to consistently bypass it. Although requests are continuously re-evaluated, the verification mechanism itself typically remains unchanged. As shown by the simulation, IZTA instead evolves the verifier through iterative optimization, suggesting that adaptive verification policies are feasible, at least computationally. IZTA improves the verifier itself. As seen through the success of the simulation, this may be possible in real-world situations.

Mutations generate candidates for a new, more robust verification policy, while selection retains only the policies that perform better. In principle, this allows the verifier itself to

adapt alongside evolving threats rather than relying on a single static policy throughout its lifetime. Although this behavior has only been demonstrated within the simulation, it illustrates how immune-inspired optimization could potentially strengthen future Zero Trust systems.

Robustness Through Diversity: Maintaining high diversity during the early stages of optimization may be one of the most important aspects of an evolving ZTA policy. High diversity allows multiple candidate solutions to be evaluated simultaneously.

This does not imply that diversity should remain high indefinitely. As demonstrated in the baseline simulation, diversity naturally decreases as the verifier converges toward a strong solution. The challenge for future implementations of IZTA would therefore be balancing continued exploration with stable exploitation of high-performing verification policies.

This may also help reduce the likelihood of becoming trapped in a suboptimal local optimum. The policy being optimized may correspond to a local optimum rather than the global optimum, while the global optimum is never reached because the system is stuck optimizing the worse option. With high diversity, multiple relative maximums can be evaluated, increasing the likelihood that higher-quality solutions are discovered.

If future versions of IZTA incorporate evolving threat representations, maintaining diversity during adaptation may allow the verifier to respond more effectively to changing threats. When a threat changes, the local and global optimum may change, so with multiple candidates all searching, even the biggest changes in the threat may be optimized for and counteracted.

Built-in Contingency: The logic behind the selection mechanism in IZTA has a built-in contingency measure. If no clone performs better than the parent, then the parent is retained instead of a subpar clone being selected.

This works as a fallback because the verifier can never get worse (as seen in the selected clone being strictly non-negative in **Figure 1**), it can only get better or stay the same,

provided the attack doesn't evolve. This mirrors established security policy, a new change is only deployed if it's also beneficial.

If all the candidate policies were worse, then they would never be deployed; the initial policy would be kept.

5. Lessons Learned & Future Work

5.1 Future Improvements

While IZTA is a functional simulation that demonstrates emergent affinity maturation behavior, it is far from a complete cybersecurity system. By implementing new features, the simulation can become more accurate to biology and more useful to cybersecurity.

The first improvements that can be made are to mitigate the limitations that I enumerated earlier. The next improvements would be in data analysis, to implement parameter sweeps to gain a fuller understanding of how each parameter affects the system (also possibly revealing more emergent behaviors).

The biggest change that I could make would be to implement dynamic antigens. This would let us see how affinity maturation combats an evolving pathogen. Additionally, this would be further proof-of-concept that adaptive ZTA could counteract evolving threats, or at least it would refute it; either way it would provide essential information.

Other changes would include, but are not limited to: multiple antigens, more, better differentiation, better lethal mutations, more data analysis metrics, more realistic biology overall, and real cybersecurity datasets.

5.2 Technical Lessons Learned

There were multiple lessons I learned related to creating the model and analyzing data while working on this project.

Parameter Balance: I've spent many hours researching extremely specific aspects of the immune system so I could have accurate parameters for my baseline. For example, I looked

into how long a single occurrence of SHM takes, how many iterations of SHM may happen, and how long a B cell spends doing SHM, just to arrive at 50 for the number of iterations.

Initially I didn't even have lethal mutations, only after researching and realizing that many mutations are downright fatal to cells did I implement it. When I first did, the lethal mutation rate was very high (50%), but I determined that it wasn't accurate to the immune system, and decreased it to 35%.

Every parameter required balancing biological realism with computational tractability. Increasing realism often introduced additional complexity, while excessive simplification risked losing the behavior being modeled.

More Iterations ≠ Better Results: Initially I had designed the simulation to run for 100 iterations so that SHM would have ample time to work, but I severely underestimated the effectiveness of the process. In the baseline run, iteration 40 is when the plateau of parent affinity becomes more pronounced. After convergence, additional iterations have a minuscule benefit on parent affinity.

Future versions of IZTA could have the iteration end point be variable and dependent on whether convergence is reached to make sure that no resources are wasted.

The Importance of Visualization: Originally, I wasn't planning on creating graphs and analyzing them; I was just going to analyze based on the raw data, but I quickly realized that doing that was not going to work. Before graphing, the raw data provided a general idea but any patterns in them were virtually impossible to see. But after graphing, I noticed so many new data patterns such as: iteration 16, phase transitions & emergence, diversity vs optimization, and convergence behaviors. For complex and dense simulations such as IZTA, visualization is a necessity.

5.3 Broader Lessons Learned

IZTA demonstrates that biological systems can provide useful design principles even when their mechanisms are heavily abstracted. The immune system does not defend the body through a fixed list of responses; it continuously adapts through variation and selection.

This idea can be translated into cybersecurity and used to provide defenses for next-generation threats.

Biological systems represent billions of years of evolutionary optimization. Studying these systems, allows us to extract these useful principles and use them for many other domains. Immunology is one of the closest analogues to cybersecurity in nature and specifically biology. This project has reinforced my view that immunology has significant potential to inspire innovations in cybersecurity.

6. Conclusion

6.1 Final Statements

Immunological Zero Trust Architecture is an abstract simulation inspired by affinity maturation facilitated through somatic hypermutation. IZTA is mainly a proof of concept that affinity maturation can work computationally rather than an enterprise cybersecurity solution.

Through the analysis of the data provided by running the simulation, I found many interesting occurrences. One of the biggest findings was of emergence, specifically that emergence still occurs in my abstracted, segmented, and artificial immune system. Only a small portion of the immune system was actually simulated, and of that part, every component was heavily simplified. I was pleasantly surprised when I noticed emergent behaviors despite the simulation being barebones compared to the actual immune system. This suggests that biological, especially immunological, principles can inform adaptive cybersecurity.

As cybersecurity threats become increasingly dynamic, an equally adaptive solution is needed to counteract these new and evolving threats. The immune system provides the inspiration for just that.

6.2 Closing Personal Note

This project initially began as a way for me to combine my passion, the immune system, and my hobby, cybersecurity, but I recognize that this project has surpassed being just a way to combine them. I've become genuinely passionate about the immune system in cybersecurity now, and since I recently learned (while finishing this write-up) about the field of adaptive cybersecurity, this project may have helped me find a field I want to explore, although, like the simulation itself, I'm still exploring that possibility rather than having converged on it.

Learning what my project was aiming to show, that immunological inspirations may create the next-generation of cybersecurity defenses, already has an established research community behind it and even its own multibillion dollar field (and independently reaching

it) was incredibly vindicating. This project no longer felt like it was exploring an idea that may never come into fruition, rather, it felt like I was contributing to a larger research base.

This project started in early April, through just some barebones Java files on GitHub and throughout the past couple months, my code has gone through countless (well GitHub commits count it) changes through countless hours. Even just graphing the data in google sheets took hours. Longest of all, was this manuscript, weeks of writing and reiterating to make sure that this manuscript turns out to be even better than the simulation. Looking back, the project itself followed a process not unlike the one it models: repeated variation, evaluation, selection, and refinement. But I'm happy to say that all this time was worth it. I wouldn't even count this as sacrificing my summer because this project is made of passion and is extremely enjoyable.

I plan to continue working on and adding to this simulation and write-up the entirety of my senior year.

Thank you for taking the time to read my work. I hope IZTA demonstrates the potential of immunologically inspired cybersecurity. - Aryan Gupta

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Data and Code Availability

The core source code, simulation parameters, and logged baseline metrics generated during this study are openly available in the Appendix of this manuscript.

Competing Interests

The author declares that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

AI Use Statement

Although sparingly, AI assistance was used by the author. AI was only used to help with certain Java code concepts the author was not familiar with, and occasionally used to commit in GitHub. The Author did not use AI in the writing of this manuscript at all.

Appendix A: Baseline Parameter Index

Table 2

Baseline Parameter Values

Parameter	Value	Explanation
Dimension	100	Dimension of the antigen & B-cell receptor feature vector
Seed	42	The initial number for the pseudorandom number generator to ensure deterministic outcomes
Iterations	50	How many iterations the simulation runs
Initial population size	100	Number of randomly generated B-cell candidates. The highest-affinity cell is selected as the initial parent
Initial population variability	0.1	A higher value makes more unique parents more likely (0 to 1)
Number of clones	25	How many clones a parent produces

Mutation magnitude	0.05	Standard deviation of the Gaussian mutation applied to receptor entries. Larger values produce larger mutations on average.
Mutation chance	0.15	Probability that an individual receptor entry undergoes mutation.
Lethal mutation chance	0.35	The chance that a B cell dies due to its mutations
Differentiation chance	0.5	Chance that a cell in any iteration will differentiate, given that it meets other criteria
Plasma threshold	0.7	The affinity threshold for a cell to differentiate into a plasma cell
Memory threshold	0.5	The affinity threshold for a cell to differentiate into a memory cell
Plasma start iteration	15	First iteration during which plasma cell differentiation is permitted
Memory start iteration	35	First iteration during which memory cell differentiation is permitted (plasma cell production also stops)

Appendix B: Iteration-Logged Parameters Index

Table 3

All metrics logged, per iteration

Logged Parameter	**Explanation**
Basic Metrics	Metrics relating directly to affinity
Parent affinity	Affinity of the iteration's progenitor cell
Best clone affinity	Affinity of the best clone; excludes parent
Selected affinity	Affinity of the selected cell; includes parent
Mean affinity	Mean affinity of the entire iteration
Δ Best affinity	The change in affinity between the previous and current best clone

Δ Mean affinity	The change in affinity between the previous and current mean
Best clone affinity (relative)	Affinity of the best clone normalized against the parent
Selected affinity (relative)	Affinity of the selected cell normalized against the parent
Mean affinity (relative)	Mean affinity of the iteration normalized against the parent
Mutation Metrics	Metrics relating directly to mutations
Clones evaluated	Amount of clones assessed; equivalent to total amount of clones in an iteration
Successful mutations	Amount of clones that didn't encounter a lethal mutation
Lethal mutations	Amount of clones that did encounter a lethal mutation
Clones survived	Amount of clones that survived mutation; equivalent to successful mutations
Clones died	Amount of clones that didn't survive mutation; equivalent to lethal mutations
Differentiation Metrics	Metrics relating directly to differentiation
Plasma cell count	Amount of plasma cells
Memory cell count	Amount of memory cells
Plasma mean affinity	Mean affinity of plasma cells
Memory mean affinity	Mean affinity of memory cells
Population Diversity Metrics	Metrics measuring population diversity
Affinity range	The difference between the highest and lowest affinity cell
Affinity standard deviation	The standard deviation between each cell

Appendix C: Artifacts & Data Availability

Source code & Commit History on Github.

Baseline System Data.